



SYDNEY TECHNICAL HIGH SCHOOL

2013

HIGHER SCHOOL CERTIFICATE

TRIAL EXAMINATION

Mathematics

Extension 1

General Instructions

- Reading time – 5 minutes.
- Working Time – 2 hours.
- Write using a blue or black pen.
- Board approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- Show all necessary working in Questions 11 – 14.
- Begin each question on a new page.
- Write your name and your teachers name on the booklet and your multiple choice answer sheet.

Total marks (70)

Section I

10 marks

- Attempt questions 1 – 10.
- Answer on the multiple choice answer sheet provided.
- Allow about 15 minutes for this section.

Section II

60 marks

- Attempt questions 11 – 14
- Answer in the booklet provided and show all necessary working.
- Start a new page for each question and clearly label it.
- Allow about 1 hour 45 minutes for this section.
- Marks are shown beside each question

Section 1**Total marks - 10**

1. The smallest positive value of x for which $\tan(2x) = 1$ is

A. 0

B. $\frac{\pi}{8}$

C. $\frac{\pi}{4}$

D. $\frac{\pi}{2}$

2. The inverse of the function $f(x) = e^{2x+3}$ is

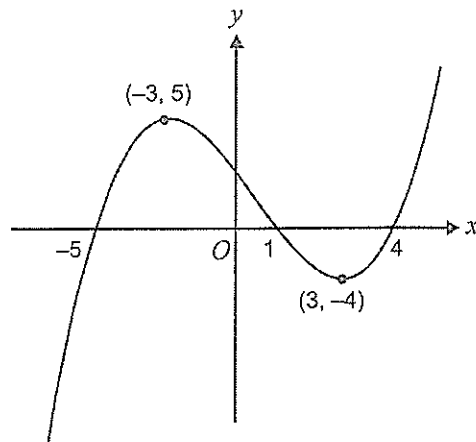
A. $f^{-1}(x) = e^{-2x-3}$

B. $f^{-1}(x) = e^{\frac{x-3}{2}}$

C. $f^{-1}(x) = \log_e(\sqrt{x}) - \frac{3}{2}$

D. $f^{-1}(x) = -\log_e(2x - 3)$

3.



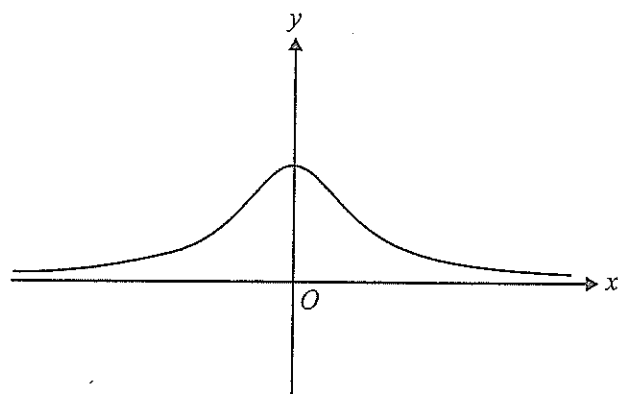
For the graph $y = f(x)$ shown above, $f'(x)$ is negative when

- A. $-3 < x < 3$
- B. $-3 \leq x \leq 3$
- C. $x < -3$ or $x > 3$
- D. $x \leq -3$ or $x \geq 3$

4. The solutions to the equation $e^{4x} - 5e^{2x} + 4 = 0$ are

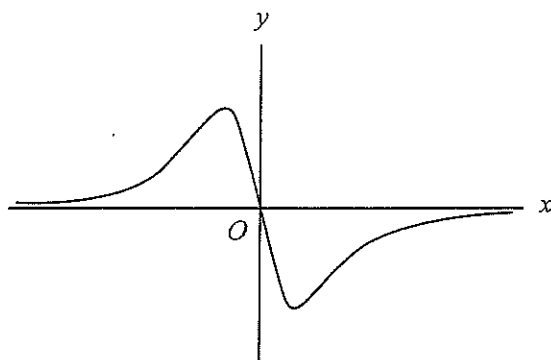
- A. 1 and 4
- B. -4 and -1
- C. $-\log_e 2$, 0 , $\log_e 2$
- D. 0 , $\log_e 2$

5. The graph of a function f is shown below

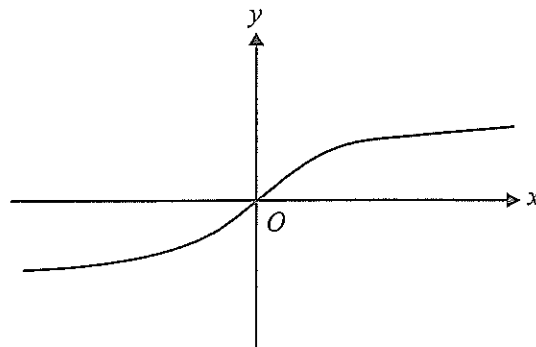


The graph of a primitive function of f could be

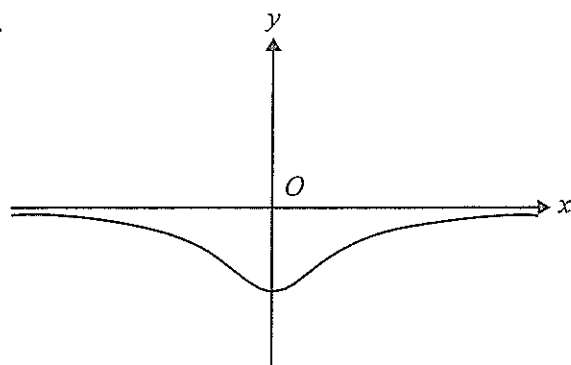
A.



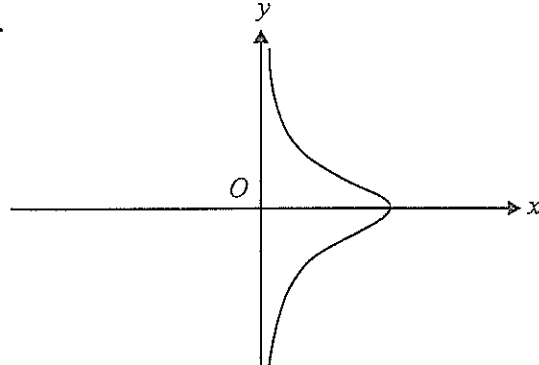
B.



C.



D.



6. The derivative of $\log_e (2f(x))$ with respect to x is

A. $\frac{f'(x)}{f(x)}$

B. $2 \frac{f'(x)}{f(x)}$

C. $\frac{f'(x)}{2f(x)}$

D. $\log_e (2f'(x))$

7. The normal to the curve with equation $y = x^{\frac{3}{2}} + x$ at the point (4,12) is parallel to the straight line with equation

A. $4x = y$

B. $4y + x = 7$

C. $y = \frac{x}{4} + 1$

D. $x - 4y = -5$

8. The function with rule $f(x) = -3 \sin \left(\frac{\pi x}{5} \right)$ has period

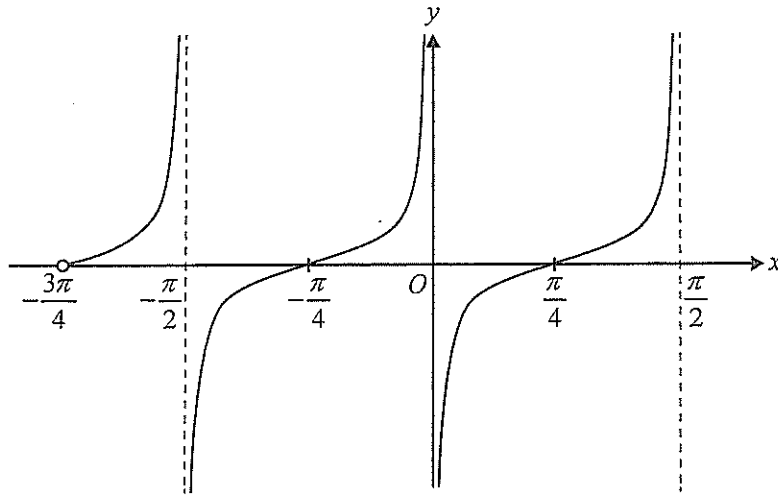
A. 3

B. 5

C. 10

D. $\frac{\pi}{5}$

9. A section of the graph of f is shown below:



The equation of f could be

- A. $f(x) = \tan x$
- B. $f(x) = \tan(x - \frac{\pi}{4})$
- C. $f(x) = \tan[2(x - \frac{\pi}{4})]$
- D. $f(x) = \tan[2(x - \frac{\pi}{2})]$

10. The equation of the chord of contact of the tangents to the parabola $x^2 = 8y$ from the point $(3, -2)$ is;

- A. $3x - 4y + 8 = 0$
- B. $3x - 8y + 16 = 0$
- C. $3x - 8y - 8 = 0$
- D. $3x - 4y + 16 = 0$

Section 2

Total marks - 60

Answer all questions starting each question on a new side of paper with your name and question number at the top of the page

Question 11 (15 marks)

- A. Find the coordinates of the point P which divides the interval from A(-3,6) to B(12, -4) in the ratio of -2:3 2
- B. Find the value $\cos 105^\circ$ in simplest exact form with a rational denominator. 3
- C. Solve the inequality $\frac{2x+1}{x-1} \geq 3$ and graph your solution on a number line 3
- D. Find $\lim_{x \rightarrow 0} \frac{\sin 6x}{7x}$ 1
- E. Use the substitution $u = t + 1$ or otherwise to evaluate $\int_0^1 \frac{t}{\sqrt{t+1}} dt$ 3
(Leave your answer in exact form)
- F. Find the acute angle, to the nearest degree, between the lines $y = 3x + 1$ and $y = -x + 6$ 3

Question 12 (15 marks) (Start a new page)

- A. (i) Show that the equation of the tangent at $T(-2t, t^2)$ on the parabola $y = \frac{1}{4}x^2$ is given by $y + tx + t^2 = 0$ 2
- (ii) The point $M(x, y)$ is the midpoint of the interval TA where A is the x intercept of the equation of the tangent at T. Find the equation of the locus of M as T moves on the parabola. 3
- B. Find $\int \frac{dx}{4+x^2}$ 1
- C. Given $f(x) = \sin^{-1} 2x$
- (i) Write down the domain and range of $f(x)$ 2
- (ii) Sketch the curve 1
- D. A spherical balloon is expanding so that its volume $V \text{ mm}^3$ increases at a constant rate of $72 \text{ mm}^3/\text{second}$. What is the rate of increase of its surface area $A \text{ mm}^2$ when the radius is 12mm. 3
- E. Use mathematical induction to prove that $n^3 + (n+1)^3 + (n+2)^3$ is divisible by 9 for all positive integers n 3

Question 13 (15 marks) (Start a new page)

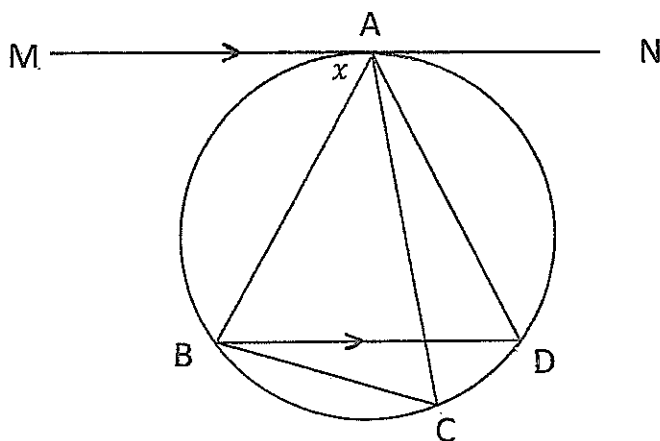
A. A particle moves in a straight line and at time t seconds, its distance $x\text{ cm}$ from a fixed point is given by $x = 1 + \frac{1}{2} \cos 2t$

(i) Show that the motion of the particle is simple harmonic by expressing $\ddot{x} = -n^2(x - A)$ 1

(ii) State the period of its motion 1

(iii) Find the displacement of the particle from the origin when it is at rest, and determine its amplitude. 2

B.



ABC is a triangle inscribed in a circle. MAN is a tangent to the circle at A. BD is a chord of the circle such that $BD \parallel MN$. Let $\angle MAB = x$ 3
 Copy diagram onto your answer sheet.
 Show that CA bisects $\angle BCD$.

C. Newton's law of cooling states that the rate of change of the temperature T of a body at any time t is proportional to the difference in temperature T of the body and the temperature m of the surrounding medium ie: $\frac{dT}{dt} = k(T - m)$ where k is a constant.

(i) Show that $T = m + Ae^{kt}$ where A is a constant, satisfies this equation 1

(ii) If the temperature of the surrounding air is 40°C and the temperature of the body drops from 170°C to 105°C in 45 minutes, find the temperature of the body in another 90 minutes (nearest whole degree)
[Find k correct to 3 decimal places] 3

(iii) Find the time taken for the temperature of the body to drop to 80°C (to the nearest minute) 2

D. Find $\int \cos^2 2x \, dx$ 2

Question 14 (15 marks) (Start a new page)

A. (i) Prove $\frac{d^2x}{dt^2} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$ where v denotes velocity 2

(ii) The acceleration of a particle moving in a straight line is given by $\ddot{x} = -2e^{-x}$ where x is the displacement from the origin. Initially the particle is at the origin with velocity 2m/s.

α . Prove that $v^2 = 4e^{-x}$ 2

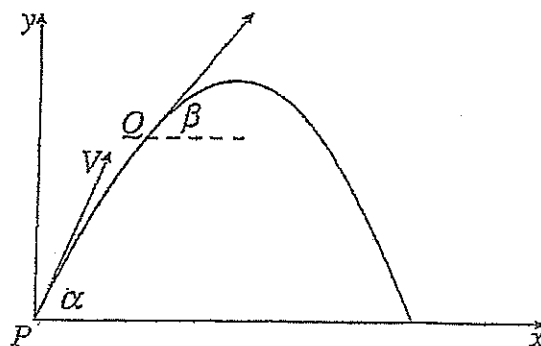
β . Describe the subsequent motion of the particle with reference to its speed and direction 2

B. $P(x)$ is a monic polynomial of degree 3. $P(x)$ has a quadratic factor of $x^2 - 1$ and when $P(x)$ is divided by $x - 2$, the remainder is -9. Form an equation for $P(x)$ and hence solve $P(x) = 0$ 2

C. A particle is projected from a point P on horizontal ground, with initial Speed V m/s at an angle of elevation α to the horizontal. Its equations of motion are $\ddot{x} = 0$ and $\ddot{y} = -g$. The horizontal and vertical components of velocity and displacement of the particle at any time t are given by

$$\frac{dx}{dt} = V \cos \alpha \text{ and } \frac{dy}{dt} = V \sin \alpha - gt$$

$$x = Vt \cos \alpha \text{ and } y = Vt \sin \alpha - \frac{1}{2} gt^2 \text{ (do not prove these)}$$



(i) Show that the time of the flight of the particle is given 2
by $t = \frac{2V\sin\alpha}{g}$

(ii) The particle reaches a point Q , as shown, where the direction 1
of the flight makes an angle β with the horizontal. Show that

$$\tan\beta = \frac{V\sin\alpha - gt}{V\cos\alpha}$$

(iii) Hence show that the time taken to travel from P to Q is 2

$$\frac{V\sin(\alpha-\beta)}{g\cos\beta} \text{ seconds}$$

(iv) Consider the case where $\beta = \frac{\alpha}{2}$. If the time taken to travel 2
from P to Q is one third of the total time of the flight, find the value of α .

Student Name: _____

Teacher Name: _____

2013 STHS Extension 1 Trial Solutions

Section I

1. B 2. C 3. A 4. D 5. B

6. A 7. B 8. C 9. C 10. A

Section II

11. a) x_1, y_1 x_2, y_2
 $A(-3, 6)$ $B(12, -4)$
 $k = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-4 - 6}{12 - (-3)} = \frac{-10}{15} = -\frac{2}{3}$
 $x = \frac{m x_2 + n x_1}{m + n}$
 $x = \frac{-2 \cdot 12 + 3 \cdot (-3)}{-2 + 3} = -33$
 $y = \frac{-2 \cdot (-4) + 3 \cdot 6}{-2 + 3} = 26$
 $(-33, 26)$ ②

b) $\cos 105^\circ$
 $\cos(60^\circ + 45^\circ)$
 $\cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ$ ①
 $\frac{1}{2} \times \frac{\sqrt{2}}{2} = \frac{\sqrt{2}}{4}$ ①
 $\frac{1 - \sqrt{3} \times \sqrt{2}}{2\sqrt{2} \times \sqrt{2}} = \frac{\sqrt{2} - \sqrt{6}}{4}$ ①

c) $\frac{2x+1}{x-1} \geq 3$
 $x \neq 1$

$2x+1 = 3x-3$
 $4 = x$
 $\leftarrow x \quad \rightarrow$
 $+1 < x \leq 4$ ②

d) $\lim_{x \rightarrow 0} \frac{\sin 6x}{6x} \times \frac{1}{6}$
 $\frac{6}{7} \times \lim_{x \rightarrow 0} \frac{\sin 6x}{6x}$
 $\frac{6}{7} \times 1$
 $= \frac{6}{7}$ ①

e) $\int_0^1 \frac{t}{t+1} dt$
 $u = t+1$
 $du = dt$
 $t=0, u=1, t=1, u=2$
 $\Rightarrow \int_1^2 \frac{u-1}{u} du$ ①
 $\int_1^2 (1 - \frac{1}{u}) du$
 $[\frac{2}{3} u^{\frac{3}{2}} - 2u^{\frac{1}{2}}]_1^2$ ①
 $\frac{2}{3} \times 2^{\frac{3}{2}} - 2\sqrt{2} - (\frac{2}{3} - 2)$
 $\frac{2}{3} \times 2^{\frac{3}{2}} - 2\sqrt{2} + 1\frac{1}{3}$ ①

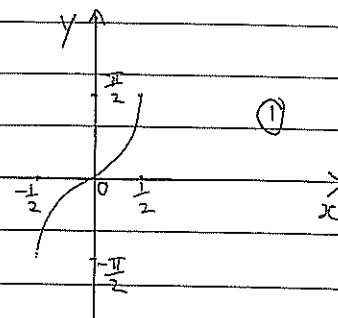
Student Name: _____

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f) $y = 3x+1$ $y = -x+6$ 12. $y = \frac{1}{4}x^2$
 $m_1 = 3$ $m_2 = -1$ $\frac{dy}{dx} = \frac{1}{2}x$
 $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$ ①
 $= \left| \frac{3 - (-1)}{1 + (-3)} \right|$ ①
 $= \left| \frac{4}{-2} \right| = 2$
 $\therefore \theta = 63^\circ$ ①

g) A is when $y=0$ b) $\int \frac{dx}{4+x^2}$
 $x^2 + 4 = 0$
 $x = -2$
 $A(-2, 0)$ Now M ①
 $\frac{1}{2} \tan^{-1}(\frac{x}{2}) + C$ ①

is midpoint of $A(-2, 0)$ c) $f(x) = \sin^{-1} 2x$
 $T(-2, 2)$ Domain: $-1 \leq 2x \leq 1$
 $\Rightarrow M(-\frac{3}{2}, \frac{1}{2})$ ①
 $x = -\frac{3}{2}, y = \frac{1}{2}$
 $2x = -3t$ $y = \frac{t^2}{2}$
 $t = \frac{2x}{3}$ $y = \frac{4x^2}{9} \times \frac{1}{2}$
 $2x^2 = 9y$ ①



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d) $\frac{dV}{dt} = 72$, find $\frac{dA}{dt}$

when $r = 12$

$$V = \frac{4}{3}\pi r^3 \quad A = 4\pi r^2$$

$$\frac{dV}{dr} = 4\pi r^2 \quad \frac{dA}{dr} = 8\pi r \quad (1)$$

$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt} \quad (1)$$

$$= \frac{dA}{dr} \times \left(\frac{dr}{dV} \times \frac{dV}{dt} \right)$$

$$= 8\pi r \times \left(\frac{1}{4\pi r^2} \times 72 \right)$$

$$= 2 \times 12 \times \frac{1}{12} \times 72$$

$$= 12 \text{ m}^2/\text{sec} \quad (1)$$

e) Step 1

Show result is true for $n=1$

ie: $n^3 + (n+1)^3 + (n+2)^3$ becomes

$$1^3 + (1+1)^3 + (1+2)^3$$

$$1 + 8 + 27$$

$$= 36 \text{ which divides by } \checkmark$$

Step 2

Assume result is true for $n=k$

$$\text{ie: } k^3 + (k+1)^3 + (k+2)^3 = M \text{ (M integral)}$$

Step 3

Show result is true for $n=k+1$

$$(k+1)^3 + (k+2)^3 + (k+3)^3$$

$$= (k+3)^3 + 9M - k^3 \text{ from step 2} \quad (2)$$

$$= k^3 + 9k^2 + 9k + 27 + 9M - k^3$$

$$= 9k^2 + 27k + 27 + 9M$$

$$= 9[k^2 + 3k + 3 + M] \text{ is integral since}$$

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M and k are integral (1) for correct setting out

Step 4

Since result is true for $n=1$, it must also be true for $n=1+1=2$, $n=2+1=3$... and hence for all positive integral values of n .

13. i) $x = 1 + \frac{1}{2} \cos 2t$

ii) Period is $\frac{2\pi}{n}$

SHM if $\ddot{x} = -n^2(x-A)$

$n=2$ (1)

$$\dot{x} = -\sin 2t$$

$\therefore \text{period} = \pi \text{ second}$

$$\ddot{x} = -2 \cos 2t \text{ but from } x,$$

$$\cos 2t = 2(x-1)$$

$\therefore \ddot{x} = -4(x-1)$ is of required form (1)

iii) At rest when

$$\dot{x} = 0$$

b) Join CD

$$\angle MAB = \angle ABD = x$$

ie: $-\sin 2t = 0$ when

$$2t = 0, \pi, 2\pi, \dots$$

$$t = 0, \frac{\pi}{2}, \pi, \dots$$

sub. any into x

$$x = 1 + \frac{1}{2} \cos 0 = \frac{3}{2} \text{ at rest}$$

$$\text{Amplitude} = \frac{1}{2} \quad (1)$$

(1) (Alternate angles $MN \parallel BD$)

$$\angle ABD = \angle ACD = x$$

(Angles in same segment)

$$\angle MAB = \angle ACB = x$$

(1) (Alternate Segment Theorem)

$$\therefore CA \text{ bisects } \angle BCD$$

$$\text{as } \angle ACB = \angle ACD = x$$

(1)

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c) i) $T = m + Ae^{kt}$
 $\frac{dT}{dt} = kAe^{kt}$ ① but
 from $T = m + Ae^{kt}$

$T - m = Ae^{kt}$ ②
 substituting ② into ①

$\frac{dT}{dt} = k(T - m)$ as req'd ①

cii) $m = 40^\circ$
 when $t = 0, T = 170$
 $t = 45, T = 105$

$170 = 40 + Ae^{k \cdot 0}$
 $\therefore A = 130$ so ①

$T = 40 + 130e^{kt}$
 $105 = 40 + 130e^{k \cdot 45}$

$65 = 130e^{45k}$
 $\frac{1}{2} = e^{45k}$

$\ln \frac{1}{2} = 45k$ ①

$k = -0.015$ Now

when $t = 135$

$T = 40 + 130e^{-0.015 \times 135}$

$T = 57^\circ$ ①

aii) $T = 40 + 130e^{-0.015t}$

Find t when $T = 80$

$80 = 40 + 130e^{-0.015t}$

$40 = 130e^{-0.015t}$ ①

$\ln \frac{4}{13} = -0.015t$

$t = 79$ minutes ①

14. ci) $\frac{d^2x}{dt^2} = \frac{d}{dt} \left(\frac{dx}{dt} \right)$

$= \frac{d}{dt} (V)$

$= \frac{dV}{dt}$

$= \frac{dV}{dx} \times \frac{dx}{dt}$

$= V \frac{dV}{dx}$ since $V = \frac{dx}{dt}$

①

$\frac{d}{dV} \left(\frac{1}{2} V^2 \right) \cdot \frac{dV}{dx}$
 $= \frac{d}{dx} \left(\frac{1}{2} V^2 \right)$ ①

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cii) a) From ci), sub

$\frac{d}{dx} \left(\frac{1}{2} V^2 \right)$ for $\ddot{x}$

$\frac{d}{dx} \left(\frac{1}{2} V^2 \right) = -2e^{-x}$

$\int dx \quad \int dx$

$\frac{1}{2} V^2 = 2e^{-x} + C$ ①

when $x = 0, V = 2$

$\frac{1}{2} \times 2^2 = 2e^0 + C$

$2 = 2 + C$

$\therefore C = 0$

$\therefore V^2 = 4e^{-x}$ ①

b) As x increases

from 0 it

moves to the

right or in a

positive direction. ①

As $x \rightarrow \infty, V \rightarrow 0$

so it is slowing

down but never

reaches 0. ①

b) Let $P(x) = (x^2 - 1)(x - 2)$

$\Rightarrow P(2) = (4 - 1)(2 - 2) = -9$ ①

$6 - 3x = -9$

$-3x = -15$

$x = 5$

$\therefore P(x) = (x - 1)(x + 1)(x - 5)$

$x = 1, -1, 5$ ①

c) i) Solve $y = 0$

$V \sin \alpha - \frac{1}{2} g t^2 = 0$

$t(V \sin \alpha - \frac{1}{2} g t) = 0$ ①

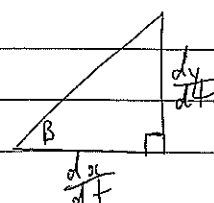
$V \sin \alpha = \frac{1}{2} g t$

$\frac{2V \sin \alpha}{g}$ or $0 = t$

$\Rightarrow$ Time of flight is

$\frac{2V \sin \alpha}{g}$ ①

cii)



$\tan \beta = \frac{dy/dt}{dx/dt}$

$= \frac{V \sin \alpha - g t}{V \cos \alpha}$ ①

$$\text{iii) } \tan \beta = \frac{V \sin \alpha - gt}{V \cos \alpha}$$

$$\frac{\sin \beta}{\cos \beta} = \frac{V \sin \alpha - gt}{V \cos \alpha}$$

Now solve for t

$$V \sin \beta \cos \alpha = V \sin \alpha \cos \beta - \cos \beta gt$$

$$\cos \beta gt = V(\sin \alpha \cos \beta - \sin \beta \cos \alpha) \quad (1)$$

$$= V \sin(\alpha - \beta)$$

$$\text{so } t = \frac{V \sin(\alpha - \beta)}{g \cos \beta} \quad (1)$$

civ) If $\beta = \frac{\alpha}{2}$ then time from P to Q is

$$t = \frac{V \sin \frac{\alpha}{2}}{g \cos \frac{\alpha}{2}} = \frac{V \tan \frac{\alpha}{2}}{g} \text{ and this is } \frac{1}{3}$$

of total flight time

$$\Rightarrow \frac{V \tan \frac{\alpha}{2}}{g} = \frac{1}{3} \times \frac{2V \sin \alpha}{g}$$

$$3V \tan \frac{\alpha}{2} = 2V \times \frac{2 \tan \frac{\alpha}{2}}{1 + \tan^2 \frac{\alpha}{2}} \quad (+ \text{ results}) \quad (1)$$

$$3 + 3 \tan^2 \frac{\alpha}{2} = 4$$

$$3 \tan^2 \frac{\alpha}{2} = 1$$

$$\tan^2 \frac{\alpha}{2} = \frac{1}{3}$$

$$\tan \frac{\alpha}{2} = \frac{1}{\sqrt{3}} \quad (\alpha \text{ is acute})$$

$$\frac{\alpha}{2} = \frac{\pi}{6}$$

$$\therefore \alpha = \frac{\pi}{3} \quad (1)$$